CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B.Tech (CL) Examination

December 2010

CL 201 Surveying I

Date: 26.11.10 , Friday **Time: 10.30 a.m to 1.30 p.m** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION- I

Q1 (a) Describe the process of measuring magnetic bearing of a line by theodolite. 05

(b) Derive the formula for calculating offsets from chords produced for setting out a simple circular curve. 05

(c) What are the methods of Plane Table Survey? Describe any one of them with neat sketch. 05

Q2 (a) Enlist the elements of Simple Circular curve and describe any one of them. 02

(b) How soundings are located ? Enlist names of nine methods of location sounding. 03

(c) The following consecutive readings were taken with a level and 4 m leveling staff on continuously sloping ground at common interval of 30 m 05

0.855(on A), 1.545, 2.335, 3.115, 3.825, 0.455, 1.380, 2.055, 2.855, 3.455, 0.585, 1.015, 1.850, 2.755, 3.845 (on B). RL of point is 380.500 m.

Enter this readings in a level book page and determine gradient of line AB. Adopt rise and fall method also apply the usual checks.

OR

Q2 (a) Explain principle of ecosounding. 02

(b) What are the elements of simple circular curve? Give their relationships. 03

(c) A road at formation level is 6 m wide and has side slope of 2:1. The road is to have a constant RL of 200.00 m, ground is leveled across the center line of the road. The following observations were made : 05

Chaninage in meters	0	20	40	60	80	100
Surface levels along Center line of the road	204.6	203.0	200.8	201.6	202.0	200.2

Estimate volume of earth

Q3 (a) Give a brief note on : Errors in Plane Table Survey 03

(b) Define the following terms : latitude and departure, face left and face right 02

(c) An incomplete traverse table is obtained as below. Calculate the length of line DA and bearing of line AB. 05

Line	Length (m)	Bearing
AB	100.00	(?)
BC	80.5	140° 30'
CD	60.0	220° 30'
DA	(?)	310° 15'

OR

Q3 (a) Define : Bench Mark 01

(b) Describe the method for setting out of a building. 04

(c) Balance the given traverse ABCD with the help of Transit Rule **05**

Line	Latitude	Departure
AB	+ 123.35	+ 35.68
BC	+ 93.82	+ 205.86
CD	- 177.44	+ 70.11
DA	- 39.21	- 312.25

SECTION – II

Q4 **(a)** With usual notations, derive the prismoidal formula to find the volume. **05**

(b) Give a brief note on different instruments used for taking soundings. **05**

(c) What is Total Station? Describe briefly salient features of total station. **05**

Q5 **(a)** What is the principle of Plane Table Surveying? **02**

(b) What are temporary adjustments of dumpy level ? How it is done ? **03**

(c) The following records refer to an operation involving reciprocal leveling **05**

Instrument at	Staff reading on		Remarks
	A	B	
A	1.155	2.595	Distance AB = 500 m
B	0.985	2.415	RL of A = 525.500 m

Find : 1. The true RL of B

2. Combined correction for curvature and refraction
3. Whether line of collimation is inclined upwards or downwards
4. Collimation error per m

OR

Q5 **(a)** Derive an expression for side widths and areas of a two level section. **03**

(b) Explain the procedure for setting out a small culvert. **03**

(c) Calculate the ordinates at 10 m distances for a circular curve having a long chord of 80 m and versed sine of 4 m **04**

Q6 **(a)** What is the basic difference between a venire theodolite and an electronic theodolite ? **03**

(b) Write short note on : Slope Rail **03**

(c) Enlist various characteristics of contour lines; support your answer with suitable sketches. **04**

OR

Q6 **(a)** Define : Contour line, Contour Interval, Horizontal Equivalent **03**

(b) What do you understand by Electronic Distance Measurement ? **03**

(c) What do you understand by orientation in plane table surveying ? Why it is required? **04**